

Efficient State Transition Algorithm With Guaranteed Optimality

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Abstract—The state transition algorithm (STA), as an intelligent optimization method grounded in constructivist learning, has been demonstrated to be highly effective in solving complex optimization problems. However, the standard STA suffers from slow convergence, particularly in the later stages when dealing with flat landscapes. Additionally, users are required to set the maximum number of iterations based on intuition. To address these issues, an enhanced STA with guaranteed optimality is introduced. This improvement involves three key components. First, novel translation transformations (TTs), inspired by predictive modeling, are developed to generate a broader set of candidate solutions by leveraging historical data. Second, adaptive parameter control strategies are incorporated to accelerate convergence. Finally, a dedicated termination condition is designed to ensure that the algorithm converges at the optimal solution, analogous to the zero gradient condition in mathematical programming. The comprehensive experimental results validate the effectiveness and superiority of the proposed method. The source codes for ESTA and EXSTA will be publicly available at <https://github.com/tiezhongyu2005/ESTA>

Index Terms—Constructivist learning, guaranteed optimality, intelligent optimization, state transition algorithm (STA), termination condition.

I. INTRODUCTION

MODERN systems engineering problems often involve complex optimization tasks spanning multiple stages of the system lifecycle, including design, analysis, and deployment. These problems are inherently characterized by nonlinearity, nonconvexity, high dimensionality, and uncertainty, thereby necessitating optimization methods that are both efficient and reliable to ensure satisfactory system performance. Intelligent optimization algorithms have been widely adopted as practical tools for solving complex optimization problems. However, most existing approaches, particularly those based on swarm intelligence and imitation learning, lack rigorous theoretical guarantees in terms of optimality, convergence, and controllability, thereby undermining their reliability in critical systems engineering applications.

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The state transition algorithm (STA) is a constructivist learning-based intelligent optimization approach developed in recent years [1], [2], [3]. In STA, each solution to an optimization problem is metaphorically treated as a state, and its iterative update is correspondingly defined as a state transition. Unlike the majority of swarm intelligence optimization algorithms based on imitation learning and behavioral cloning [4], the STA is inspired by constructivist learning [5], which acquires the essential knowledge and constructs the necessary components, including globality, optimality, rapidity, convergence, and controllability, aiming to find a global or near-global optimal solution as quickly as possible. In standard STA [6], a set of diverse state transformation operators is specifically designed to generate candidate solutions to achieve this goal. Specifically, the expansion transformation (ET) facilitates global search to ensure globality, while the rotation transformation (RT) supports local search to enhance local optimality. Additionally, translation transformation (TT) and axesion transformation (AT) are employed for heuristic search to improve rapidity. Furthermore, appropriate update strategies are implemented to ensure convergence. Moreover, each state transformation operator generates a regular neighborhood with an adjustable size, providing controllability. Due to its ease of understanding, convenience of use, as well as powerful search ability, it has been successfully applied in a wide range of practical engineering problems [7], [8], [9], [10].

However, on the one hand, the standard STA tends to exhibit slow convergence in the later stages, particularly when dealing with flat landscapes, due to the use of fixed parameter settings. Many efforts have been made to address this issue. In [11], all state transformation factors are decreased periodically to enhance local exploitation in the later stages. In [1], an optimized strategy for parameter selection was proposed to accelerate convergence; however, it necessitates some time to initiate the acceleration process. In [12], gradient information is incorporated to accelerate the search process in the later stages; nevertheless, it requires the function to be differentiable. The question of how to integrate appropriate mechanisms to accelerate convergence in STA for a general function requires further investigation.

On the other hand, the maximum number of iterations in standard STA is manually specified as a termination criterion, typically based on prior experience, which is common practice in intelligent optimization algorithms. As is well known, for an intelligent optimization algorithm, besides generating new candidates and updating incumbents, the termination condition is also a crucial component. In mathematical programming

with continuously differentiable objective function, a widely used termination condition is the zero gradient condition, as it guarantees the optimality due to Fermat's theorem. However, for the majority of existing intelligent optimization algorithms, it is challenging to design an automatic termination criterion to guarantee the optimality [13], [14], [15]. The simplest and most widely used termination criterion in intelligent optimization algorithms is a predefined number of function evaluations (or a maximum number of iterations). However, determining this value is highly problem-dependent and often relies on empirical tuning. This presents a tuning dilemma: an overly low value may fail to reach the optimum, whereas an excessively high one can lead to unnecessary computational overhead. What is more, using the maximum number of iterations as a stopping criterion can be detrimental, and how to set a good termination condition in intelligent optimization algorithms still remains unresolved and requires further investigation [16]. Nevertheless, the issue of how to design an appropriate termination criterion to ensure optimality is often overlooked in studies on intelligent optimization, including previous research on STA. In earlier versions of STA, the termination criterion was also based on a predefined maximum number of iterations, which must be specified in advance and typically relies on expert knowledge. Notably, the intrinsic characteristics in STA, particularly the RT that enables the search within a hypersphere of a specified radius, make it possible to set a dedicated termination condition that can guarantee the optimality automatically.

To address the aforementioned issues, this study aims to develop a set of efficient STA variants that not only achieve faster convergence on flat landscapes but also guarantee optimality through an integrated automatic termination mechanism, which is a crucial yet often overlooked aspect in most existing studies on intelligent optimization algorithms. The main contributions and innovations presented in this article are summarized as follows.

- 1) Novel TTs based on predictive modeling are developed to generate a larger pool of potential candidates and accelerate the convergence process.
- 2) Parameter control strategies are incorporated into STA to promote faster convergence.
- 3) Specific termination criteria are designed to provide theoretical guarantees of optimality.

The rest of this article is organized as follows. Section II reviews the related work, including fundamental principles of STA and commonly used termination conditions in intelligent optimization. Section III provides the details of the proposed STA, including the novel state transformations, parameter control strategies, and dedicated termination criteria. In Section IV, comprehensive experimental results and analyses are given to demonstrate the effectiveness and superiority of the proposed STA. Finally, Section V concludes this article.

II. RELATED WORK

In this study, the following general unconstrained optimization problem is considered:

$$\min_{\mathbf{x} \in \Omega} f(\mathbf{x}) \quad (1)$$

where $f(\mathbf{x})$ denotes a continuous function with a lower bound and $\Omega \subseteq \mathbb{R}^n$ is a set that is both closed and compact.

A. Basics of STA

The STA is an intelligent optimization method grounded in constructivist learning, based on states and state transitions. Given the current state $\mathbf{s}_k$, the unified form for generating the candidate state $\mathbf{s}_{k+1}$ in STA is formulated as follows:

$$\mathbf{s}_{k+1} = A_k \mathbf{s}_k + B_k \mathbf{u}_k \quad (2)$$

where $\mathbf{s}_k$ represents a state corresponding to a candidate solution of the optimization problem for the k th iteration; $\mathbf{u}_k$ can be treated either as a function of current and historical states or independently defined; and A_k and B_k serve as state transition matrices, which, together with the update rule in (2), constitute the core state transformation operators.

Drawing on state-space representation as inspiration, four specialized state transformation operators are elaborately designed to facilitate the solution search for the optimization problem.

- 1) RT is

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \alpha R_r \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|_2} \quad (3)$$

where $\alpha > 0$ is the rotation factor, $R_r \in \mathbb{R}$ is a uniformly distributed random variable defined over the interval $[-1, 1]$, and $\mathbf{u}_k \in \mathbb{R}^n$ is a random vector whose entries are uniformly distributed random variables within the interval $[-1, 1]$. This RT enables the search within a hypersphere with the maximum radius α (the detailed proof can be found in [11]), as illustrated in Fig. 1(a) and (b).

- 2) TT is

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \beta R_t \frac{\mathbf{s}_k - \mathbf{s}_{k-1}}{\|\mathbf{s}_k - \mathbf{s}_{k-1}\|_2} \quad (4)$$

where $\beta > 0$ is the translation factor and $R_t \in \mathbb{R}$ is a uniformly distributed random variable over the interval $[0, 1]$. The TT enables the search along the direction of $\mathbf{s}_k - \mathbf{s}_{k-1}$, with a maximum step length of β , as illustrated in Fig. 1(c) and (d).

- 3) ET is

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \gamma R_e \mathbf{s}_k \quad (5)$$

where $\gamma > 0$ is the expansion factor, and $R_e \in \mathbb{R}^{n \times n}$ is a random diagonal matrix whose entries follow the normal (Gaussian) distribution, which expands the entries of $\mathbf{s}_k$ to $(-\infty, +\infty)$ to enable a search over the whole space, as illustrated in Fig. 1(e) and (f).

- 4) AT is

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \delta R_a \mathbf{s}_k \quad (6)$$

where $\delta > 0$ is the axesion factor and $R_a \in \mathbb{R}^{n \times n}$ is a diagonal matrix in which only one diagonal entry is nonzero and is drawn from a normal (Gaussian) distribution. This AT is designed to conduct a one-dimensional (1-D) search along the coordinate axes, thereby enhancing the search capability in a single dimension, as illustrated in Fig. 1(g) and (h).

By leveraging state transformation operators, a sampling technique, and an update strategy, the core procedure of the basic STA can be described in the following pseudocode [1]:

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1: repeat
2:   if  $\alpha < \alpha_{\min}$  then
3:      $\alpha \leftarrow \alpha_{\max}$ 
4:   end if
5:    $\text{Best} \leftarrow \text{expansion}(*, \text{Best}, \text{SE}, \beta, \gamma)$ 
6:    $\text{Best} \leftarrow \text{rotation}(*, \text{Best}, \text{SE}, \alpha, \beta)$ 
7:    $\text{Best} \leftarrow \text{axesion}(*, \text{Best}, \text{SE}, \beta, \delta)$ 
8:    $\alpha \leftarrow \alpha/(fc)$ 
9: until the termination criterion is satisfied

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where SE is the sample size, and the termination criterion is usually the maximum number of function evaluations or iterations. It should be emphasized that the TT operator is only executed when a new best solution is found in another state transformation, and the update strategy $f(\text{Best}_{k+1}) \leq f(\text{Best}_k)$ is used to update the incumbent best solution. For more details, please refer to [1].

B. Principle Behind the STA

The primary goal of STA is to find a global or near-global optimal solution to the optimization problem as quickly as possible. As a constructivist learning-based intelligent optimization method, STA possesses the following five core structural components.

- 1) *Globality*: The capability to explore the entire search space.
- 2) *Optimality*: The guarantee of achieving an optimal solution.
- 3) *Rapidity*: Reducing computational complexity as much as possible.
- 4) *Convergence*: Guaranteeing the convergence of the generated sequence.
- 5) *Controllability*: Flexibly managing the search space.

The globality mainly lies in the ET operator. Given that R_e is a random diagonal matrix with entries following a Gaussian distribution, by using the ET with a nonzero vector s_k and sufficiently large expansion factor γ , the entries in s_{k+1} can be expanded to the range of $(-\infty, +\infty)$, from the perspective of probability theory.

The optimality mainly lies in the RT operator. It can be proved that $\|s_{k+1} - s_k\| \leq \alpha$ by using the RT [6], [11]. That is to say, with a sufficiently large sample size SE and sufficiently small rotation factor $\alpha_{\min}$, if no better candidate s_{k+1} is found, the incumbent s_k can be regarded as an optimal solution in the mathematical sense, achieved with a prescribed level of precision $\alpha_{\min}$.

The rapidity lies in several aspects: 1) the sampling technique: based on the incumbent solution, the candidates generated by each state transformation operator form a regular neighborhood, from which representative candidates can be selected by sampling, avoiding exhaustive enumeration; 2) the alternate strategy: the alternative use of different transformation operators facilitates escape from regions around local

optima; and 3) efficient implementation; for example, the time complexity of the RT implementation has been reduced from $O(n^2)$ to $O(n)$ [11].

The convergence mainly lies in the update rule: $f(\text{Best}_{k+1}) \leq f(\text{Best}_k)$. Since the sequence $\{f(\text{Best}_k)\}$ is monotonically decreasing and $f(\text{Best})$ is bounded below, the sequence $\{f(\text{Best}_k)\}$ converges.

The controllability mainly lies in the transformation factors α, β, γ , and δ . These transformation factors can regulate the search space to a desired region with a specific size. For example, the RT enables the search within a hypersphere of maximal radius α .

Theorem 1: If the sample size SE is sufficiently large and the rotation factor α is sufficiently small at the later stages, the sequence $\{f(\text{Best}_k)\}$ generated by the STA can converge to a minimum, i.e.,

$$\lim_{k \rightarrow \infty} f(\text{Best}_k) = f(x^*) \quad (7)$$

where x^* is a minimum.

Proof: First, under the update rule $f(\text{Best}_{k+1}) \leq f(\text{Best}_k)$ in STA, the sequence $f(\text{Best}_k)$ is monotonically nonincreasing and bounded below. Therefore, by the monotone convergence theorem, the sequence $f(\text{Best}_k)$ converges.

Second, since the sample size SE is sufficiently large and the rotation factor α is sufficiently small, it is not difficult to derive that $f(\text{Best}_k) \leq f(s_{k+1}) \forall s_{k+1} \in \mathcal{S}$, where $\mathcal{S} = \{s \mid \|s - \text{Best}_k\| \leq \alpha_{\min}\}$ is the neighborhood of Best_k , that is to say, $f(x^*) = \lim_{k \rightarrow \infty} f(\text{Best}_k)$ can be considered as a minimum with solution accuracy $\alpha_{\min}$.

To sum up, the sequence $\{f(\text{Best}_k)\}$ generated by the STA can converge to a minimum with certain solution accuracy. $\square$

Remark 1: The “guaranteed optimality” in this study means that the proposed algorithms converge to a minimum, either a local minimum or a global minimum for the unconstrained optimization problem. A key drawback of many intelligent optimization algorithms is their tendency to converge prematurely to a stable point, which may not represent the true optimum [17].

Remark 2: The sample size depends on the dimension of the optimization problem. Based on our empirical experience, the recommended sample size SE is set to $2n$ for low dimensions, $5n$ for medium dimensions, and $10n$ for high dimensions.

C. Analysis of the Termination Criteria in Intelligent Optimization

In intelligent optimization, most studies focus on generating candidate solutions and updating incumbents to balance global exploration and local exploitation. Termination criteria are often overlooked, with a predefined number of function evaluations or iterations typically used as the stopping condition. However, termination criteria play a crucial role in solving real-world optimization problems. If they are set too small, the algorithm may fail to find the optimal solution; on the contrary, if set too large, the algorithm may result in a waste of computing resources. As a result, extensive trial and error is required, which is not feasible in a time-constrained environment. In the academic community, the indicators of

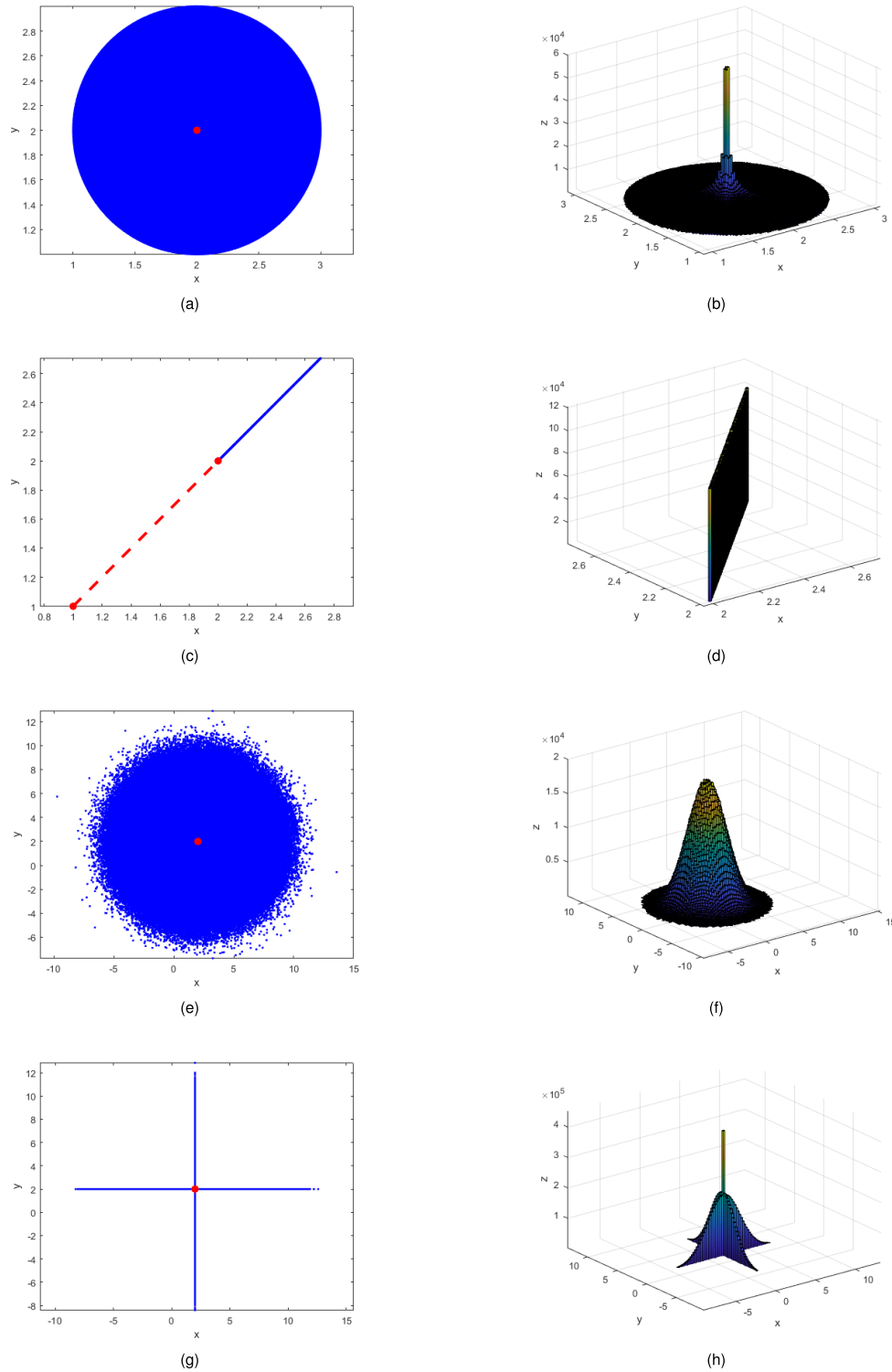


Fig. 1. Visualization of two-dimensional (2-D) plots and three-dimensional (3-D) histograms of state transformations. (a) 2-D plot of rotation. (b) 3-D histogram of rotation. (c) 2-D plot of translation. (d) 3-D histogram of translation. (e) 2-D plot of expansion. (f) 3-D histogram of expansion. (g) 2-D plot of axes. (h) 3-D histogram of axes.

termination criteria can be summarized into the following three categories [18], [19], [20], [21].

1) Resource Indicators:

- a) max iterations;
- b) max function evaluations;
- c) max CPU time.

2) Progress Indicators:

- a) accuracy, i.e., the distance to the optimum is less than a given tolerance;
- b) hitting a bound, i.e., the best found objective function value hits a given bound;
- c) K -iterations, i.e., there is no obvious improvement after K consecutive generations or iterations.

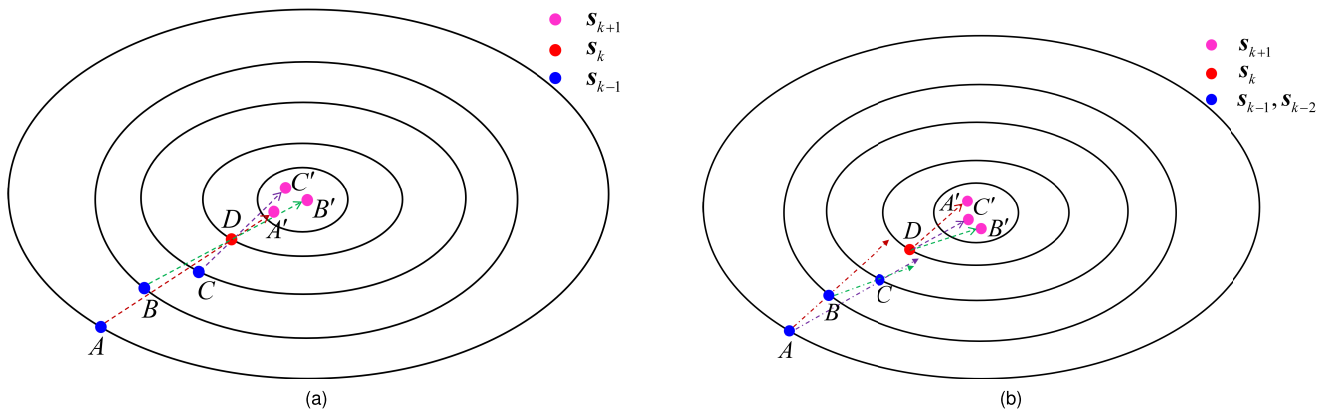


Fig. 2. Visualization and comparison of two novel TTs. (a) Novel TT based on the first-order predictive model. (b) Novel TT based on the second-order predictive model.

3) Characteristic Indicators:

- convergence;
- diversity;
- clustering;
- statistic.

Although various termination criteria have been proposed in intelligent optimization, determining an appropriate termination criterion for most algorithms remains an open problem. This is because the best solution found may not be strictly optimal in a mathematical sense when computational resources are exhausted, progress has stagnated, or certain stopping conditions are met. Fortunately, as discussed earlier, STA possesses intrinsic properties that enable the design of a termination criterion with theoretical guarantees of optimality.

III. PROPOSED EFFICIENT STA

Considering that the standard STA exhibits slow convergence in later stages, particularly on functions with flat landscapes, novel state transformations and parameter adaptation strategies are proposed to accelerate the convergence in this section.

A. Novel TT Based on Predictive Modeling

In STA, the TT is employed as a heuristic search mechanism to generate more potential solutions, and it can search along a line from the previous best solution to the current best solution. This is intuitive, as the current best solution improves upon the previous one. From the perspective of regression analysis, the TT in standard STA is a first-order predictive model with normalized scale, but only the last best solution is utilized. To make sufficient utilization of historical best solutions, in this part, novel TTs based on first- and second-order predictive models are proposed.

First, let us recall the widely used autoregressive integrated moving average (ARIMA; p, d , and q) model [22]

$$\mathbf{x}_t = \mathbf{c} + \sum_{i=1}^p \phi_i \mathbf{x}_{t-i} + \boldsymbol{\epsilon}_t + \sum_{i=1}^q \theta_i \boldsymbol{\epsilon}_{t-i} \quad (8)$$

where $\mathbf{x}_t$ is the actual value; $\boldsymbol{\epsilon}_t$ is the Gaussian white noise; p and q are referred to as the autoregressive and moving average orders, respectively; and d is the degree of differencing.

To make sure that the series can converge, new TTs based on the first- and second-order predictive models are proposed as follows:

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \beta \hat{\mathbf{R}}_t (\mathbf{s}_k - \mathbf{s}_{k-1}) \quad (9a)$$

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \beta \hat{\mathbf{R}}_t (\mathbf{s}_{k-1} - \mathbf{s}_{k-2}) \quad (9b)$$

where $\mathbf{s}_k$ represents the current state and $\mathbf{s}_{k-1}$ and $\mathbf{s}_{k-2}$ are historical states. $\hat{\mathbf{R}}_t \in \mathbb{R}$ is a uniformly distributed random variable over the interval $[-1, 1]$. Furthermore, to reflect the influence of the degree of differencing d , $\mathbf{s}_{k-1}$ and $\mathbf{s}_{k-2}$ are chosen randomly from the historical best archive.

The differences between the first- and second-order predictive models are illustrated in Fig. 2. With the same previous best solutions A, B , and C , the current best solution D and the next candidate solutions A', B' , and C' are different using different predictive models. In the first-order predictive model, A, D , and A' are in the same line, the same to B, D , and B' , and C, D , and C' . In the second-order predictive model, the line AB is parallel to line DA' , the same to line BC and line DB' , and line AC and line DC' . Although the newly generated solutions differ, the convergence rate is governed by the same term $\beta \hat{\mathbf{R}}_t$, as shown in (9).

B. New Expansion and AT for Supplementation

In the standard STA, the ET operator performs global search, whereas the AT operator conducts heuristic search. Although these two transformation operators perform well in most scenarios, their effectiveness diminishes in an extreme case. As shown in Fig. 3, the global point B (where $x = 2$) is far away from the local point A (where $x = 0.2$). If the incumbent's best solution $\mathbf{s}_k$ is around the local point A and the state transformation factor is set at 1, it is extremely difficult to jump out of the neighborhood since the probability $P(\mathbf{s}_{k+1} \geq 0.8) \approx 0.13\%$.

To resolve the abovementioned issue, new expansion and AT are designed as follows:

$$\text{expansion : } \mathbf{s}_{k+1} = \mathbf{s}_k + \gamma \mathbf{R}_e \mathbf{u}_k$$

$$\text{axesion : } \mathbf{s}_{k+1} = \mathbf{s}_k + \delta \mathbf{R}_a \mathbf{u}_k. \quad (10)$$

Here, $\mathbf{u}_k = [1, \dots, 1]^T$ is the all-ones vector.

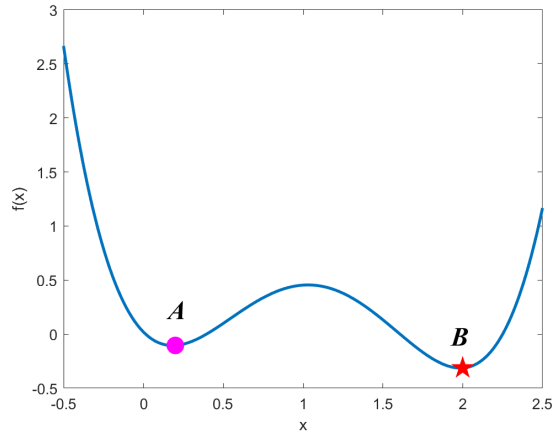


Fig. 3. Illustration of an extreme case.

Remark 3: For the extreme case, by using the new AT, the probability to jump out of the local point A is $P(s_{k+1} \geq 1) \approx 21\%$. Given that the original expansion and AT perform well in most cases, these new transformations are therefore combined with the original ones.

C. Proposed Parameter Control Strategy

Parameter setting is a key issue in intelligent optimization algorithms, as their performance strongly depends on parameter values. Parameter setting can be broadly categorized into parameter tuning and parameter control. In parameter tuning, suitable parameter values are selected in advance and kept fixed throughout the optimization process. In contrast, parameter control dynamically adjusts parameter values during the search to improve performance and adaptability. Since identifying optimal parameter values is often challenging and time-consuming and no single parameter configuration is universally effective across different problems, parameter control becomes essential, particularly for complex optimization tasks.

Parameter setting is also important in STA, as the state transformation factors have been shown to significantly influence its performance. In the standard STA [6], the translation factor β , the expansion factor γ , and the axesion factor δ are kept constant, while the rotation factor α decreases exponentially from 1 to 10^{-4} with base 2 in a periodic manner. In dynamic STA (DaSTA) [11], all state transformation factors decrease exponentially from 1 to 10^{-4} with base 2 in a periodic manner to enhance local exploitation in the later stages. In parameter-optimal STA (POSTA) [1], a parameter selection strategy is proposed to accelerate convergence. Specifically, parameter values are selected from the set $1, 10^{-1}, 10^{-2}, \dots, 10^{-8}$ and kept constant for a predefined period to ensure full utilization.

1) *Intuitive Parameter Adaptation Strategy:* First, we focus on two key state transformation factors: the rotation factor α and the expansion factor γ , which correspond to local search and global search, respectively. The central idea is to adapt these parameters according to changes in the incumbent solution. As is well known, larger parameter values promote global exploration, whereas smaller values favor local exploitation. Moreover, global exploration is typically emphasized in the early stages to explore unknown regions, while local

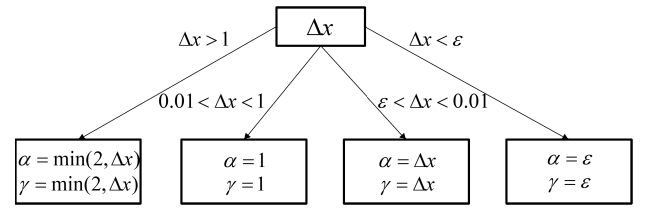


Fig. 4. Intuitive parameter adaptation strategy.

exploitation is strengthened in the later stages to refine the solution. Based on our previous studies, setting the state transformation factors to 1 provides a balanced and moderate choice. If the values are too large, redundant and ineffective candidate solutions are generated, leading to inefficiency. Conversely, if the values are too small in the early stages, the improvement of the solution becomes slow.

Based on the above analysis, an intuitive parameter adaptation strategy is proposed, as illustrated in Fig. 4, where the change between two consecutive incumbent solutions is defined as follows:

$$\Delta x = \max(|\mathbf{Best}_k - \mathbf{Best}_{k-1}|) \quad (11)$$

where $\mathbf{Best}_k$ and $\mathbf{Best}_{k-1}$ are the current best solution and the previous best solution, respectively. It utilizes the maximum variation in two incumbent solutions as the indicator to adjust the state transformation factors α and γ . ε in Fig. 4 represents the precision level of the solution, which is specified by the user.

2) *Parameter-Optimal Selection Strategy:* On the other hand, as indicated in POSTA [1], the impact of parameter on STA is predictable, as its parameter is analogous to the learning rate, also known as the step size in gradient-based optimization algorithms, as shown in the following:

$$\left. \begin{array}{l} R_f \frac{u_k}{\|u_k\|_2} \\ \hat{R}_f(s_k - s_{k-1}) \\ \dots \end{array} \right\} \Rightarrow \left. \begin{array}{l} \alpha \\ \beta \\ \dots \end{array} \right\} \Rightarrow a_k, s_{k+1} = s_k + a_k d_k$$

where d_k is analogous to the search direction in trust region methods.

Assuming that the incumbent solution is not optimal, sufficiently small parameter values allow the corresponding state transformation operator to generate improved candidate solutions. In contrast, overly large or inappropriate parameter values may lead to inefficient search.

With this in mind, the parameter selection problem can be viewed as analogous to line search in classical mathematical programming methods. For simplicity, an inexact line search strategy is adopted, as described in the following equation:

$$a^* = \underset{a_k \in \Omega}{\operatorname{argmin}} f(s_k + a_k d_k) \quad (12)$$

where $\Omega = \{2, 1, 10^{-1}, 10^{-2}, \dots, 10^{-8}\}$.

D. Proposed Efficient STA With Guaranteed Optimality

With the aforementioned novel state transformation operators and parameter control strategies, the procedure of the proposed efficient STA with guaranteed optimality is given as follows:

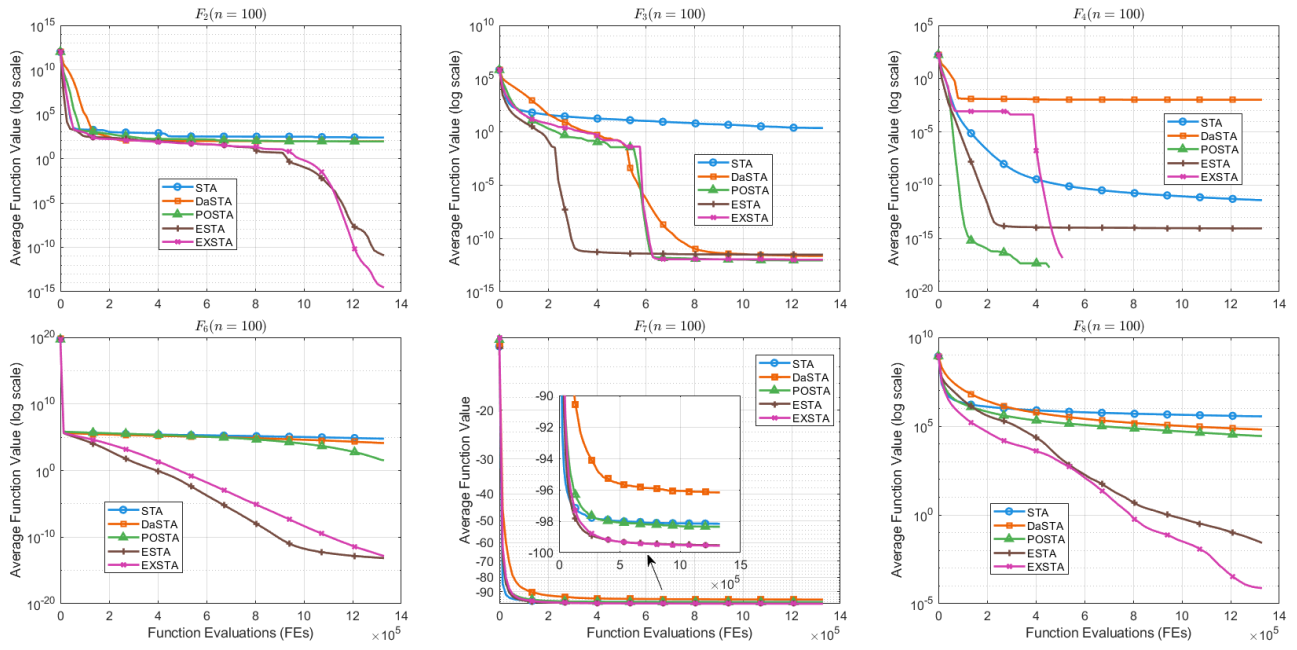


Fig. 5. Iterative curves of STA and its variants for 100-dimensional problems.

-
- 1: **repeat**
 - 2: $\text{prevfBest} \leftarrow \text{fBest}$
 - 3: $[\text{Best}, \text{fBest}, \text{Archive}] \leftarrow \text{expansion}(*, \text{Best}, \text{Archive}, \gamma)$
 - 4: $[\text{Best}, \text{fBest}, \text{Archive}] \leftarrow \text{rotation}(*, \text{Best}, \text{Archive}, \alpha)$
 - 5: $[\text{Best}, \text{fBest}, \text{Archive}] \leftarrow \text{axesion}(*, \text{Best}, \text{Archive}, \delta)$
 - 6: $[\text{Best}, \text{fBest}, \text{Archive}] \leftarrow \text{translation}(*, \text{Best}, \text{Archive}, \beta)$
 - 7: $[\alpha, \beta, \gamma, \delta] \leftarrow \text{UpdateParameter}()$
 - 8: **until** $\text{fBest} == \text{prevfBest} \ \& \ \alpha \leq \varepsilon$
-

where the Archive is to store historical best solutions and ε is the solution accuracy.

Remark 4: The designed termination condition indicates that if the rotation factor α decreases below a predefined threshold ε without any further progress ($\text{fBest} == \text{prevfBest}$), the STA will terminate automatically. By the way, to distinguish between different parameter control strategies, the proposed STA, which incorporates an intuitive parameter adaptation strategy that is simple yet efficient, is abbreviated as ESTA. In the meantime, the extended version that employs the optimal parameter selection strategy is called EXSTA.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

In this section, extensive experiments are conducted to evaluate the performance of the proposed effective STA. To avoid redundant comparisons among functions with similar characteristics and center-bias issue in evolutionary computation [23], a set of representative benchmark functions was selected, most of which are shifted, as presented in Table I. They cover a diverse range of types, including Type I: unimodal (F_1, F_9, F_{11}, F_{12}) and multimodal ($F_2 - F_8, F_{10}, F_{13} - F_{15}$); Type II: separable (F_1, F_3, F_7, F_{13}) and non-separable ($F_2, F_4 - F_6, F_8 - F_{12}, F_{14}, F_{15}$); Type III: smooth ($F_1 - F_{10}$) and unsmooth ($F_{11} - F_{15}$); and Type IV: flatten (F_2, F_7, F_8) and unflatten ($F_1, F_3 - F_6, F_9 - F_{15}$).

For a comprehensive evaluation, the proposed efficient STA is first compared with several STA variants, including the standard STA [6], DaSTA [11], and POSTA [1]. It is then compared with other representative intelligent optimization algorithms, including well-established methods such as GL25 [24], CLPSO [25], CMA-ES [26], and LSHADE [27], as well as popular methods such as GWO [28] and WOA [29], and recently proposed methods such as PGA [30] and KLA [31].

For comparisons with other optimization algorithms, the experiments are divided into two groups based on termination criteria: maximum stall iterations (maxStalls) for benchmark functions and maximum function evaluations (maxFEs) for a real-world problem. Each benchmark function is tested in 20, 30, 50, and 100 dimensions, and all algorithms are independently executed 30 times in MATLAB (R2022b) on a PC equipped with an Intel¹ Core² i7-1260P CPU @ 2.10 GHz.

For parameter settings, all competing optimization algorithms use their default parameter values. In STA and its variants, the sample size SE is set to 30. For the proposed termination criterion, convergence is assumed when $\text{prevBest} - \text{fBest} \leq \text{eps}$ (a constant in MATLAB), indicating that fBest remains unchanged. The solution accuracy threshold ε is set to 10^{-8} . The archive size is set to 100 in ESTA and EXSTA. When applying the optimal parameter selection strategy, the hyperparameters are set to be the same as those in POSTA.

B. Comparison Among Different Predictive Models

As described in Section III, three types of predictive models are considered: the first-order model, the second-order model, and their hybrid. In this section, a comparative study is conducted to examine the differences among these models.

¹Registered trademark.

²Trademarked.

TABLE I
USED BENCHMARK FUNCTIONS

Name	Function	Range	x^*	$F(x^*)$
shifted Sphere	$F_1(\mathbf{x}) = f_1(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_1(\mathbf{x}) = \sum_{i=1}^n x_i^2$	$[0, 2n]$	$[1, \dots, n]$	0
shifted Rosenbrock	$F_2(\mathbf{x}) = f_2(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_2(\mathbf{x}) = \sum_{i=1}^n (100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2)$	$[0, 2n]$	$[2, \dots, n+1]$	0
shifted Rastrigin	$F_3(\mathbf{x}) = f_3(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_3(\mathbf{x}) = \sum_{i=1}^n (x_i^2 - 10 \cos(2\pi x_i) + 10)$	$[0, 2n]$	$[1, \dots, n]$	0
shifted Griewank	$F_4(\mathbf{x}) = f_4(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_4(\mathbf{x}) = \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1$	$[0, 2n]$	$[1, \dots, n]$	0
shifted Ackley	$F_5(\mathbf{x}) = f_5(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_5(\mathbf{x}) = 20 + e - 20 \exp\left(-0.2 \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}\right) - \exp\left(\frac{1}{n} \sum_{i=1}^n \cos(2\pi x_i)\right)$	$[0, 2n]$	$[1, \dots, n]$	0
shifted Zakharov	$F_6(\mathbf{x}) = f_6(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_6(\mathbf{x}) = \sum_{i=1}^n x_i^2 + \left(\sum_{i=1}^n 0.5ix_i\right)^2 + \left(\sum_{i=1}^n 0.5ix_i\right)^4$	$[0, 2n]$	$[1, \dots, n]$	0
Michalewicz	$F_7(\mathbf{x}) = -\sum_{i=1}^n \sin(x_i) \sin\left(\frac{ix_i^2}{\pi}\right)^{20}$	$[0, \pi]$	unknown	unknown
Trid	$F_8(\mathbf{x}) = \sum_{i=1}^n (x_i - 1)^2 - \sum_{i=2}^n x_i x_{i-1} + \frac{n(n+4)(n-1)}{6}$	$[0, n^2]$	$x_i^* = i(n+1-i)$	0
shifted Schwefel 1.2	$F_9(\mathbf{x}) = f_9(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_9(\mathbf{x}) = \sum_{i=1}^n \left(\sum_{j=1}^i x_j\right)^2$	$[0, 2n]$	$[1, \dots, n]$	0
shifted Schwefel 2.4	$F_{10}(\mathbf{x}) = f_{10}(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_{10}(\mathbf{x}) = \sum_{i=1}^n (x_i - 1)^2 + (x_1 - x_i^2)^2$	$[0, 2n]$	$[2, \dots, n+1]$	0
shifted Schwefel 2.21	$F_{11}(\mathbf{x}) = f_{11}(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_{11}(\mathbf{x}) = \max\{ x_i , 1 \leq i \leq n\}$	$[0, 2n]$	$[1, \dots, n]$	0
shifted Schwefel 2.22	$F_{12}(\mathbf{x}) = f_{12}(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_{12}(\mathbf{x}) = \sum_{i=1}^n x_i + \prod_{i=1}^n x_i $	$[0, 2n]$	$[1, \dots, n]$	0
Schwefel 2.26	$F_{13}(\mathbf{x}) = \sum_{i=1}^n -x_i \sin(\sqrt{ x_i }) + 418.9829n$	$[-500, 500]$	$[420.9687, \dots]$	0
shifted Penalized1	$F_{14}(\mathbf{x}) = f_{14}(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_{14}(\mathbf{x}) = \frac{\pi}{n} \left\{ 10 \sin(\pi y_1) + \sum_{i=1}^{n-1} (y_i - 1)^2 [1 + 10 \sin^2(\pi y_{i+1})] \right.$ $\left. + (y_n - 1)^2 \right\} + \sum_{i=1}^n u(x_i, 10, 100, 4), y_i = 1 + \frac{x_i + 1}{4}$ $u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m, & x_i > a \\ 0, & -a \leq x_i \leq a \\ k(-x_i - a)^m, & x_i < -a \end{cases}$	$[0, 2n]$	$[0, \dots, n-1]$	0
shifted Penalized2	$F_{15}(\mathbf{x}) = f_{15}(\mathbf{x} - \mathbf{s}), \quad \mathbf{s} = 1 : n$ $f_{15}(\mathbf{x}) = 0.1 \left\{ \sin^2(3\pi x_1) + \sum_{i=1}^{n-1} (x_i - 1)^2 [1 + \sin^2(3\pi x_{i+1})] \right\} +$ $(x_n - 1)^2 [1 + \sin^2(2\pi x_n)] + \sum_{i=1}^n u(x_i, 5, 100, 4)$	$[0, 2n]$	$[2, \dots, n+1]$	0

Different ESTA variants based on these predictive models, all equipped with the proposed termination criterion, are compared, and the statistical results are reported in Table II. The mean (denoted by **m**) and standard deviation (denoted by **sd**) of the objective function values (abbreviated as **ObjVal**) are used to evaluate the search performance and stability of the algorithms. Meanwhile, the mean gradient norm (abbreviated as **GradNorm**) is used to assess optimality.

From the statistical results in Table II, it can be observed that all ESTA variants based on predictive models, equipped with the proposed termination criterion, achieve optimality, as the gradient norm is on the order of 10^{-4} or smaller (a commonly adopted tolerance in engineering applications). Most of these variants are able to locate the global optimum with satisfactory accuracy, except for the Griewank function in 20 dimensions and the Michalewicz function, for which the global optimum

TABLE II
COMPARISON OF ESTA VARIANTS BASED ON DIFFERENT PREDICTIVE MODELS WITH THE PROPOSED TERMINATION CRITERION

Fcn	Dim	First-order model		Second-order model		Hybrid model	
		ObjVal (m $\pm$ sd)	GradNorm (m)	ObjVal (m $\pm$ sd)	GradNorm (m)	ObjVal (m $\pm$ sd)	GradNorm (m)
F_1	20	$1.74 \times 10^{-15} \pm 3.42 \times 10^{-16}$	8.30×10^{-8}	$1.67 \times 10^{-15} \pm 2.76 \times 10^{-16}$	8.14×10^{-8}	$1.73 \times 10^{-15} \pm 4.59 \times 10^{-16}$	8.73×10^{-8}
	30	$3.05 \times 10^{-15} \pm 4.65 \times 10^{-16}$	1.12×10^{-7}	$4.27 \times 10^{-15} \pm 1.53 \times 10^{-15}$	1.29×10^{-7}	$3.29 \times 10^{-15} \pm 6.23 \times 10^{-16}$	1.14×10^{-7}
	50	$7.40 \times 10^{-15} \pm 1.59 \times 10^{-15}$	1.71×10^{-7}	$1.05 \times 10^{-14} \pm 1.20 \times 10^{-15}$	2.04×10^{-7}	$7.08 \times 10^{-15} \pm 1.06 \times 10^{-15}$	1.68×10^{-7}
	100	$1.73 \times 10^{-14} \pm 1.74 \times 10^{-15}$	2.63×10^{-7}	$1.79 \times 10^{-14} \pm 3.04 \times 10^{-15}$	2.72×10^{-7}	$1.97 \times 10^{-14} \pm 4.76 \times 10^{-15}$	2.83×10^{-7}
F_2	20	$1.63 \times 10^{-14} \pm 7.22 \times 10^{-15}$	4.50×10^{-6}	$3.65 \times 10^{-14} \pm 1.96 \times 10^{-14}$	3.50×10^{-6}	$1.33 \times 10^{-14} \pm 2.20 \times 10^{-15}$	4.10×10^{-6}
	30	$2.78 \times 10^{-14} \pm 1.71 \times 10^{-14}$	5.26×10^{-6}	$4.82 \times 10^{-14} \pm 2.91 \times 10^{-14}$	6.84×10^{-6}	$2.21 \times 10^{-13} \pm 1.90 \times 10^{-13}$	4.56×10^{-6}
	50	$1.90 \times 10^{-13} \pm 7.75 \times 10^{-14}$	7.87×10^{-6}	$2.85 \times 10^{-13} \pm 2.29 \times 10^{-13}$	8.83×10^{-6}	$7.96 \times 10^{-14} \pm 3.84 \times 10^{-14}$	7.87×10^{-6}
	100	$4.78 \times 10^{-11} \pm 1.64 \times 10^{-11}$	2.37×10^{-5}	$8.97 \times 10^{-13} \pm 3.91 \times 10^{-13}$	1.53×10^{-5}	$1.79 \times 10^{-12} \pm 3.14 \times 10^{-13}$	1.70×10^{-5}
F_3	20	$2.84 \times 10^{-13} \pm 0$	1.64×10^{-5}	$1.98 \times 10^{-13} \pm 6.84 \times 10^{-14}$	1.39×10^{-5}	$2.80 \times 10^{-13} \pm 8.57 \times 10^{-14}$	1.64×10^{-5}
	30	$1.10 \times 10^{-12} \pm 1.51 \times 10^{-13}$	3.13×10^{-5}	$2.11 \times 10^{-12} \pm 6.99 \times 10^{-13}$	4.12×10^{-5}	$1.12 \times 10^{-12} \pm 1.15 \times 10^{-13}$	3.18×10^{-5}
	50	$4.59 \times 10^{-12} \pm 8.95 \times 10^{-13}$	6.18×10^{-5}	$3.89 \times 10^{-12} \pm 1.51 \times 10^{-12}$	5.61×10^{-5}	$2.87 \times 10^{-12} \pm 1.41 \times 10^{-12}$	5.00×10^{-5}
	100	$2.87 \times 10^{-11} \pm 3.76 \times 10^{-12}$	1.52×10^{-4}	$2.52 \times 10^{-11} \pm 7.03 \times 10^{-12}$	1.45×10^{-4}	$2.37 \times 10^{-11} \pm 5.08 \times 10^{-13}$	1.40×10^{-4}
F_4	20	$8.11 \times 10^{-3} \pm 6.86 \times 10^{-3}$	2.41×10^{-8}	$1.29 \times 10^{-2} \pm 1.44 \times 10^{-2}$	2.31×10^{-8}	$7.27 \times 10^{-2} \pm 8.60 \times 10^{-2}$	2.88×10^{-8}
	30	$4.98 \times 10^{-15} \pm 6.87 \times 10^{-16}$	3.78×10^{-8}	$1.12 \times 10^{-14} \pm 5.75 \times 10^{-15}$	4.17×10^{-8}	$1.06 \times 10^{-14} \pm 5.40 \times 10^{-15}$	3.80×10^{-8}
	50	$1.76 \times 10^{-14} \pm 6.50 \times 10^{-15}$	4.23×10^{-8}	$2.13 \times 10^{-14} \pm 5.84 \times 10^{-15}$	4.38×10^{-8}	$1.99 \times 10^{-14} \pm 1.33 \times 10^{-15}$	4.35×10^{-8}
	100	$1.13 \times 10^{-13} \pm 3.22 \times 10^{-14}$	7.49×10^{-8}	$7.90 \times 10^{-14} \pm 1.39 \times 10^{-14}$	7.61×10^{-8}	$9.27 \times 10^{-14} \pm 3.02 \times 10^{-14}$	7.06×10^{-8}
F_5	20	$2.11 \times 10^{-9} \pm 6.36 \times 10^{-10}$	2.11×10^{-4}	$2.69 \times 10^{-9} \pm 1.69 \times 10^{-10}$	2.69×10^{-4}	$2.84 \times 10^{-9} \pm 1.50 \times 10^{-10}$	2.84×10^{-4}
	30	$3.32 \times 10^{-9} \pm 7.57 \times 10^{-10}$	3.32×10^{-4}	$3.15 \times 10^{-9} \pm 5.98 \times 10^{-10}$	3.15×10^{-4}	$2.07 \times 10^{-9} \pm 3.07 \times 10^{-10}$	2.07×10^{-4}
	50	$3.12 \times 10^{-9} \pm 5.22 \times 10^{-10}$	3.12×10^{-4}	$5.73 \times 10^{-9} \pm 2.92 \times 10^{-9}$	5.73×10^{-4}	$4.50 \times 10^{-9} \pm 1.15 \times 10^{-11}$	4.50×10^{-4}
	100	$3.84 \times 10^{-9} \pm 1.06 \times 10^{-9}$	2.43×10^{-4}	$4.48 \times 10^{-9} \pm 3.35 \times 10^{-10}$	4.48×10^{-4}	$3.72 \times 10^{-9} \pm 1.45 \times 10^{-9}$	2.80×10^{-4}
F_6	20	$5.07 \times 10^{-15} \pm 5.13 \times 10^{-16}$	5.28×10^{-7}	$1.44 \times 10^{-14} \pm 8.06 \times 10^{-15}$	4.04×10^{-7}	$8.93 \times 10^{-15} \pm 5.04 \times 10^{-15}$	2.98×10^{-7}
	30	$6.40 \times 10^{-14} \pm 4.71 \times 10^{-14}$	7.30×10^{-7}	$3.55 \times 10^{-14} \pm 1.44 \times 10^{-14}$	6.06×10^{-7}	$2.35 \times 10^{-14} \pm 1.22 \times 10^{-14}$	6.72×10^{-7}
	50	$6.66 \times 10^{-14} \pm 5.15 \times 10^{-14}$	1.65×10^{-6}	$1.02 \times 10^{-13} \pm 4.75 \times 10^{-14}$	1.55×10^{-6}	$5.68 \times 10^{-14} \pm 1.54 \times 10^{-14}$	8.01×10^{-7}
	100	$1.33 \times 10^{-12} \pm 3.44 \times 10^{-13}$	2.65×10^{-6}	$7.54 \times 10^{-13} \pm 1.81 \times 10^{-13}$	6.87×10^{-6}	$7.96 \times 10^{-13} \pm 3.25 \times 10^{-13}$	9.37×10^{-6}
F_7	20	$-1.96 \times 10^1 \pm 5.40 \times 10^{-4}$	2.42×10^{-5}	$-1.96 \times 10^1 \pm 1.04 \times 10^{-2}$	1.83×10^{-5}	$-1.96 \times 10^1 \pm 7.73 \times 10^{-5}$	2.15×10^{-5}
	30	$-2.96 \times 10^1 \pm 1.02 \times 10^{-2}$	6.16×10^{-4}	$-2.96 \times 10^1 \pm 3.61 \times 10^{-3}$	5.41×10^{-5}	$-2.96 \times 10^1 \pm 5.55 \times 10^{-3}$	5.70×10^{-5}
	50	$-4.96 \times 10^1 \pm 2.66 \times 10^{-3}$	2.61×10^{-4}	$-4.96 \times 10^1 \pm 6.72 \times 10^{-3}$	2.60×10^{-4}	$-4.96 \times 10^1 \pm 1.16 \times 10^{-3}$	1.99×10^{-4}
	100	$-9.96 \times 10^1 \pm 1.46 \times 10^{-3}$	1.70×10^{-3}	$-9.96 \times 10^1 \pm 3.02 \times 10^{-3}$	1.07×10^{-3}	$-9.96 \times 10^1 \pm 1.14 \times 10^{-3}$	1.30×10^{-3}
F_8	20	$9.35 \times 10^{-10} \pm 9.83 \times 10^{-11}$	4.05×10^{-5}	$1.19 \times 10^{-9} \pm 1.04 \times 10^{-9}$	3.89×10^{-5}	$1.00 \times 10^{-9} \pm 3.13 \times 10^{-10}$	3.94×10^{-5}
	30	$6.65 \times 10^{-8} \pm 3.79 \times 10^{-8}$	1.67×10^{-4}	$2.47 \times 10^{-8} \pm 8.85 \times 10^{-9}$	1.38×10^{-4}	$3.53 \times 10^{-8} \pm 8.36 \times 10^{-9}$	1.56×10^{-4}
	50	$7.01 \times 10^{-6} \pm 3.82 \times 10^{-6}$	1.28×10^{-3}	$3.46 \times 10^{-6} \pm 1.45 \times 10^{-6}$	1.23×10^{-3}	$2.27 \times 10^{-6} \pm 1.18 \times 10^{-6}$	1.27×10^{-3}
	100	$2.74 \times 10^{-2} \pm 3.73 \times 10^{-3}$	2.74×10^{-2}	$1.23 \times 10^{-3} \pm 8.91 \times 10^{-4}$	2.93×10^{-2}	$3.28 \times 10^{-3} \pm 1.52 \times 10^{-3}$	2.52×10^{-2}
F_9	20	$7.76 \times 10^{-15} \pm 3.46 \times 10^{-15}$	2.68×10^{-7}	$8.58 \times 10^{-15} \pm 2.10 \times 10^{-15}$	1.99×10^{-7}	$7.86 \times 10^{-15} \pm 3.70 \times 10^{-15}$	1.71×10^{-7}
	30	$1.25 \times 10^{-14} \pm 1.81 \times 10^{-15}$	3.99×10^{-7}	$1.47 \times 10^{-14} \pm 2.66 \times 10^{-15}$	2.51×10^{-7}	$2.22 \times 10^{-14} \pm 1.20 \times 10^{-14}$	3.01×10^{-7}
	50	$7.55 \times 10^{-14} \pm 3.06 \times 10^{-14}$	7.56×10^{-7}	$4.02 \times 10^{-14} \pm 9.35 \times 10^{-15}$	5.90×10^{-7}	$6.74 \times 10^{-14} \pm 1.81 \times 10^{-14}$	5.80×10^{-7}
	100	$4.07 \times 10^{-13} \pm 1.56 \times 10^{-13}$	1.70×10^{-6}	$2.89 \times 10^{-13} \pm 1.59 \times 10^{-13}$	1.46×10^{-6}	$3.04 \times 10^{-13} \pm 4.32 \times 10^{-14}$	1.25×10^{-6}
F_{10}	20	$1.18 \times 10^{-15} \pm 2.26 \times 10^{-16}$	1.56×10^{-7}	$1.83 \times 10^{-15} \pm 4.11 \times 10^{-16}$	1.72×10^{-7}	$2.22 \times 10^{-15} \pm 9.92 \times 10^{-16}$	2.02×10^{-7}
	30	$4.13 \times 10^{-15} \pm 1.13 \times 10^{-15}$	2.74×10^{-7}	$3.84 \times 10^{-15} \pm 1.46 \times 10^{-15}$	2.66×10^{-7}	$5.19 \times 10^{-15} \pm 2.30 \times 10^{-15}$	2.75×10^{-7}
	50	$1.22 \times 10^{-14} \pm 6.24 \times 10^{-15}$	4.32×10^{-7}	$6.37 \times 10^{-15} \pm 3.74 \times 10^{-15}$	3.22×10^{-7}	$1.05 \times 10^{-14} \pm 8.88 \times 10^{-16}$	4.11×10^{-7}
	100	$2.57 \times 10^{-14} \pm 3.95 \times 10^{-15}$	5.73×10^{-7}	$1.75 \times 10^{-14} \pm 7.55 \times 10^{-16}$	5.07×10^{-7}	$2.28 \times 10^{-14} \pm 2.23 \times 10^{-15}$	5.47×10^{-7}

TABLE III
WILCOXON RANK-SUM TEST RESULTS OF DIFFERENT PREDICTIVE MODELS

Type	20D			30D			50D			100D		
	#+	#=	#-	#+	#=	#-	#+	#=	#-	#+	#=	#-
First-order model	5	15	0	2	15	3	2	11	7	0	12	8
Second-order model	2	16	2	3	15	2	4	12	4	6	14	0
Hybrid model	0	15	5	2	16	2	7	11	2	4	14	2

Note: #+, #=, and #- represent the count of instances where one model performs significantly better than, are statistically equivalent to, and are significantly worse than its counterparts, respectively.

is unknown. Furthermore, based on the Wilcoxon rank-sum test results in Table III, the first-order model performs better in low-dimensional problems, whereas the second-order model is more suitable for high-dimensional ones. The hybrid model, however, performs well across different dimensions and is therefore adopted in the remainder of this study.

C. Comparison With STA Variants

In this section, the proposed effective STAs are compared with STA variants. The termination condition is set to $\max FEs = 2e3n \log_2(n)$. The experimental results are shown in Tables IV and V and Fig. 5. It is observed that, with the prescribed maxFEs, the standard STA performs well on F_1 , F_3 – F_6 and F_9 – F_{15} , but not on F_2 , F_7 , and F_8 . In other words, the standard STA converges slowly on flat functions. The DaSTA enhances the standard STA in both search capability and solution accuracy on nonflat functions, but it requires

more time due to the use of probability in accepting a worse solution. Compared with standard STA, the POSTA alleviates the difficulty with flat functions and also improves solution accuracy on nonflat functions. Compared with standard STA and its other variants, it can be observed that the proposed efficient STAs exhibit a markedly improved convergence rate in the later stages, especially for the flat functions (F_2 , F_7 , and F_8), primarily attributed to the proposed new TT based on predictive modeling.

The proposed efficient STA variants achieve superior overall performance, including enhanced search capability, convergence rate, and solution accuracy. Notably, ESTA and EXSTA outperform other STA variants in all cases.

D. Comparison With Other Algorithms

In this section, the proposed efficient STAs are compared with other algorithms under two termination criteria. The first

TABLE IV
PERFORMANCE COMPARISON OF STA VARIANTS UNDER THE maxFES TERMINATION CRITERION

Fcn	Dim	STA	DaSTA	POSTA	ESTA	EXSTA
F_1	20	$8.04 \times 10^{-12} \pm 2.75 \times 10^{-12}$	$7.63 \times 10^{-18} \pm 2.86 \times 10^{-18}$	$3.86 \times 10^{-20} \pm 1.51 \times 10^{-20}$	$1.34 \times 10^{-18} \pm 7.43 \times 10^{-19}$	$1.06 \times 10^{-22} \pm 8.15 \times 10^{-23}$
	30	$1.12 \times 10^{-11} \pm 2.79 \times 10^{-12}$	$9.51 \times 10^{-18} \pm 2.81 \times 10^{-18}$	$4.60 \times 10^{-20} \pm 1.35 \times 10^{-20}$	$2.09 \times 10^{-18} \pm 8.84 \times 10^{-19}$	$7.62 \times 10^{-22} \pm 6.23 \times 10^{-22}$
	50	$1.36 \times 10^{-11} \pm 2.70 \times 10^{-12}$	$1.31 \times 10^{-17} \pm 3.84 \times 10^{-18}$	$7.47 \times 10^{-20} \pm 1.79 \times 10^{-20}$	$4.03 \times 10^{-18} \pm 1.51 \times 10^{-18}$	$1.16 \times 10^{-20} \pm 5.18 \times 10^{-21}$
	100	$1.97 \times 10^{-11} \pm 3.02 \times 10^{-12}$	$1.80 \times 10^{-17} \pm 3.41 \times 10^{-18}$	$1.13 \times 10^{-19} \pm 2.42 \times 10^{-20}$	$7.21 \times 10^{-18} \pm 1.94 \times 10^{-18}$	$3.48 \times 10^{-20} \pm 9.37 \times 10^{-21}$
F_2	20	8.57 ± 5.53	$1.28 \times 10^1 \pm 2.65$	$1.15 \times 10^1 \pm 2.94$	$2.17 \times 10^{-14} \pm 2.13 \times 10^{-14}$	$5.14 \times 10^{-16} \pm 8.22 \times 10^{-16}$
	30	$5.08 \times 10^1 \pm 3.46 \times 10^1$	$1.96 \times 10^1 \pm 3.71$	$2.05 \times 10^1 \pm 2.06$	$9.18 \times 10^{-14} \pm 1.60 \times 10^{-13}$	$5.36 \times 10^{-17} \pm 9.21 \times 10^{-17}$
	50	$8.87 \times 10^1 \pm 5.10 \times 10^1$	$3.46 \times 10^1 \pm 7.64$	$3.00 \times 10^1 \pm 1.42 \times 10^1$	$1.45 \times 10^{-13} \pm 1.27 \times 10^{-13}$	$3.81 \times 10^{-16} \pm 4.97 \times 10^{-16}$
	100	$2.29 \times 10^2 \pm 9.65 \times 10^1$	$8.59 \times 10^1 \pm 2.69 \times 10^1$	$8.26 \times 10^1 \pm 3.94 \times 10^1$	$1.21 \times 10^{-11} \pm 2.12 \times 10^{-11}$	$3.00 \times 10^{-15} \pm 6.83 \times 10^{-15}$
F_3	20	$3.14 \times 10^{-9} \pm 1.49 \times 10^{-9}$	$3.65 \times 10^{-14} \pm 2.66 \times 10^{-14}$	$3.04 \times 10^{-14} \pm 3.31 \times 10^{-14}$	$3.76 \times 10^{-14} \pm 2.79 \times 10^{-14}$	0 ± 0
	30	$8.65 \times 10^{-9} \pm 4.61 \times 10^{-9}$	$1.19 \times 10^{-13} \pm 5.85 \times 10^{-14}$	$1.14 \times 10^{-13} \pm 0$	$1.59 \times 10^{-13} \pm 6.23 \times 10^{-14}$	$1.35 \times 10^{-13} \pm 7.23 \times 10^{-14}$
	50	$3.95 \times 10^{-8} \pm 5.24 \times 10^{-8}$	$2.82 \times 10^{-13} \pm 1.30 \times 10^{-13}$	$2.10 \times 10^{-13} \pm 7.78 \times 10^{-14}$	$3.39 \times 10^{-13} \pm 1.75 \times 10^{-13}$	$1.75 \times 10^{-13} \pm 6.61 \times 10^{-14}$
	100	$2.31 \pm 8.92 \times 10^{-1}$	$2.29 \times 10^{-12} \pm 4.89 \times 10^{-13}$	$8.59 \times 10^{-13} \pm 2.32 \times 10^{-13}$	$2.99 \times 10^{-12} \pm 8.24 \times 10^{-13}$	$1.03 \times 10^{-12} \pm 2.93 \times 10^{-13}$
F_4	20	$3.97 \times 10^{-2} \pm 6.42 \times 10^{-2}$	$6.52 \times 10^{-2} \pm 6.42 \times 10^{-2}$	$2.29 \times 10^{-1} \pm 2.77 \times 10^{-1}$	$9.16 \times 10^{-3} \pm 1.50 \times 10^{-2}$	0 ± 0
	30	$2.83 \times 10^{-12} \pm 9.22 \times 10^{-13}$	$1.89 \times 10^{-2} \pm 1.88 \times 10^{-2}$	0 ± 0	$5.38 \times 10^{-16} \pm 2.38 \times 10^{-16}$	0 ± 0
	50	$3.35 \times 10^{-12} \pm 8.85 \times 10^{-13}$	$1.51 \times 10^{-2} \pm 1.74 \times 10^{-2}$	0 ± 0	$1.70 \times 10^{-15} \pm 6.61 \times 10^{-16}$	0 ± 0
	100	$3.95 \times 10^{-12} \pm 7.01 \times 10^{-13}$	$1.03 \times 10^{-2} \pm 9.24 \times 10^{-3}$	0 ± 0	$8.68 \times 10^{-15} \pm 2.68 \times 10^{-15}$	0 ± 0
F_5	20	$2.88 \times 10^{-6} \pm 3.25 \times 10^{-7}$	$3.32 \times 10^{-9} \pm 8.38 \times 10^{-10}$	$2.01 \times 10^{-10} \pm 4.20 \times 10^{-11}$	$1.12 \times 10^{-9} \pm 3.93 \times 10^{-10}$	$1.55 \times 10^{-11} \pm 7.27 \times 10^{-12}$
	30	$2.73 \times 10^{-6} \pm 4.91 \times 10^{-7}$	$3.49 \times 10^{-9} \pm 1.01 \times 10^{-9}$	$1.92 \times 10^{-10} \pm 2.93 \times 10^{-11}$	$1.18 \times 10^{-9} \pm 2.98 \times 10^{-10}$	$2.00 \times 10^{-11} \pm 7.14 \times 10^{-12}$
	50	$2.48 \times 10^{-6} \pm 2.73 \times 10^{-7}$	$5.68 \times 10^{-9} \pm 1.90 \times 10^{-9}$	$2.73 \times 10^{-10} \pm 3.63 \times 10^{-11}$	$1.21 \times 10^{-9} \pm 1.64 \times 10^{-10}$	$6.50 \times 10^{-11} \pm 1.08 \times 10^{-11}$
	100	$3.67 \times 10^{-6} \pm 5.89 \times 10^{-7}$	$2.00 \times 10^1 \pm 2.78 \times 10^{-3}$	$2.00 \times 10^1 \pm 5.31 \times 10^{-5}$	$3.82 \times 10^{-9} \pm 1.83 \times 10^{-9}$	$1.61 \times 10^{-10} \pm 5.08 \times 10^{-11}$
F_6	20	$1.70 \times 10^{-9} \pm 7.33 \times 10^{-10}$	$1.31 \times 10^{-10} \pm 1.23 \times 10^{-10}$	$3.68 \times 10^{-18} \pm 1.37 \times 10^{-18}$	$8.26 \times 10^{-17} \pm 5.54 \times 10^{-17}$	$1.04 \times 10^{-19} \pm 7.43 \times 10^{-20}$
	30	$2.20 \times 10^{-8} \pm 8.97 \times 10^{-9}$	$4.85 \times 10^{-7} \pm 3.77 \times 10^{-7}$	$3.68 \times 10^{-17} \pm 1.34 \times 10^{-17}$	$3.15 \times 10^{-16} \pm 1.29 \times 10^{-16}$	$1.58 \times 10^{-18} \pm 8.80 \times 10^{-19}$
	50	$1.31 \times 10^{-8} \pm 1.03 \times 10^{-8}$	$7.40 \times 10^{-3} \pm 6.48 \times 10^{-3}$	$2.91 \times 10^{-11} \pm 3.57 \times 10^{-11}$	$2.58 \times 10^{-15} \pm 8.29 \times 10^{-16}$	$3.88 \times 10^{-17} \pm 1.59 \times 10^{-17}$
	100	$6.17 \times 10^4 \pm 1.33 \times 10^4$	$1.33 \times 10^4 \pm 5.14 \times 10^3$	$3.33 \times 10^1 \pm 2.01 \times 10^1$	$6.92 \times 10^{-14} \pm 2.00 \times 10^{-14}$	$1.43 \times 10^{-13} \pm 8.55 \times 10^{-14}$
F_7	20	$-1.86 \times 10^1 \pm 1.03$	$-1.88 \times 10^1 \pm 9.30 \times 10^{-1}$	$-1.86 \times 10^1 \pm 1.32$	$-1.96 \times 10^1 \pm 7.44 \times 10^{-3}$	$-1.96 \times 10^1 \pm 2.63 \times 10^{-3}$
	30	$-2.89 \times 10^1 \pm 6.72 \times 10^{-1}$	$-2.79 \times 10^1 \pm 1.26$	$-2.89 \times 10^1 \pm 7.36 \times 10^{-1}$	$-2.96 \times 10^1 \pm 1.47 \times 10^{-2}$	$-2.96 \times 10^1 \pm 1.51 \times 10^{-2}$
	50	$-4.86 \times 10^1 \pm 8.59 \times 10^{-1}$	$-4.70 \times 10^1 \pm 1.55$	$-4.86 \times 10^1 \pm 8.84 \times 10^{-1}$	$-4.96 \times 10^1 \pm 1.69 \times 10^{-2}$	$-4.96 \times 10^1 \pm 1.71 \times 10^{-2}$
	100	$-9.82 \times 10^1 \pm 1.26$	$-9.62 \times 10^1 \pm 1.75$	$-9.83 \times 10^1 \pm 7.07 \times 10^{-1}$	$-9.95 \times 10^1 \pm 1.86 \times 10^{-2}$	$-9.96 \times 10^1 \pm 1.54 \times 10^{-2}$
F_8	20	1.61 ± 1.78	$2.94 \times 10^{-2} \pm 2.04 \times 10^{-2}$	$1.47 \times 10^{-2} \pm 9.11 \times 10^{-3}$	$2.96 \times 10^{-10} \pm 2.04 \times 10^{-10}$	$4.99 \times 10^{-11} \pm 4.81 \times 10^{-11}$
	30	$2.06 \times 10^2 \pm 1.77 \times 10^2$	$1.20 \times 10^1 \pm 1.01 \times 10^1$	$1.06 \times 10^1 \pm 8.79$	$8.31 \times 10^{-9} \pm 4.60 \times 10^{-9}$	$1.29 \times 10^{-9} \pm 9.33 \times 10^{-10}$
	50	$6.28 \times 10^3 \pm 4.11 \times 10^3$	$1.17 \times 10^3 \pm 8.71 \times 10^2$	$2.68 \times 10^2 \pm 2.82 \times 10^2$	$1.28 \times 10^{-6} \pm 8.25 \times 10^{-7}$	$1.13 \times 10^{-7} \pm 6.30 \times 10^{-8}$
	100	$3.57 \times 10^5 \pm 3.55 \times 10^5$	$6.49 \times 10^4 \pm 2.68 \times 10^4$	$2.78 \times 10^4 \pm 2.45 \times 10^4$	$2.67 \times 10^{-2} \pm 2.42 \times 10^{-2}$	$7.56 \times 10^{-5} \pm 4.93 \times 10^{-5}$
F_9	20	$1.08 \times 10^{-7} \pm 6.31 \times 10^{-8}$	$2.15 \times 10^{-5} \pm 1.86 \times 10^{-5}$	$4.91 \times 10^{-11} \pm 5.47 \times 10^{-11}$	$1.49 \times 10^{-16} \pm 9.39 \times 10^{-17}$	$3.75 \times 10^{-19} \pm 1.76 \times 10^{-19}$
	30	$1.15 \times 10^{-5} \pm 4.89 \times 10^{-6}$	$5.36 \times 10^{-4} \pm 2.43 \times 10^{-4}$	$6.77 \times 10^{-8} \pm 6.07 \times 10^{-8}$	$9.43 \times 10^{-16} \pm 4.56 \times 10^{-16}$	$6.32 \times 10^{-18} \pm 3.19 \times 10^{-18}$
	50	$4.52 \times 10^{-3} \pm 1.99 \times 10^{-3}$	$2.39 \times 10^{-2} \pm 9.35 \times 10^{-3}$	$1.14 \times 10^{-4} \pm 7.43 \times 10^{-5}$	$8.52 \times 10^{-15} \pm 3.24 \times 10^{-15}$	$8.06 \times 10^{-17} \pm 2.15 \times 10^{-17}$
	100	8.46 ± 2.74	3.56 ± 1.03	$1.41 \times 10^{-1} \pm 4.27 \times 10^{-2}$	$2.84 \times 10^{-13} \pm 5.21 \times 10^{-14}$	$1.50 \times 10^{-15} \pm 3.57 \times 10^{-16}$
F_{10}	20	$2.18 \times 10^{-10} \pm 1.21 \times 10^{-10}$	$4.17 \times 10^{-14} \pm 3.60 \times 10^{-14}$	$7.63 \times 10^{-19} \pm 3.09 \times 10^{-19}$	$1.03 \times 10^{-17} \pm 5.62 \times 10^{-18}$	$1.05 \times 10^{-21} \pm 9.21 \times 10^{-22}$
	30	$3.80 \times 10^{-10} \pm 1.37 \times 10^{-10}$	$6.20 \times 10^{-14} \pm 4.61 \times 10^{-14}$	$1.57 \times 10^{-18} \pm 5.88 \times 10^{-19}$	$2.29 \times 10^{-17} \pm 1.36 \times 10^{-17}$	$3.51 \times 10^{-21} \pm 2.25 \times 10^{-21}$
	50	$5.80 \times 10^{-10} \pm 1.94 \times 10^{-10}$	$6.41 \times 10^{-14} \pm 3.32 \times 10^{-14}$	$2.17 \times 10^{-18} \pm 5.75 \times 10^{-19}$	$4.10 \times 10^{-17} \pm 1.57 \times 10^{-17}$	$7.43 \times 10^{-20} \pm 2.82 \times 10^{-20}$
	100	$1.13 \times 10^{-9} \pm 2.02 \times 10^{-10}$	$1.15 \times 10^{-13} \pm 4.19 \times 10^{-14}$	$4.34 \times 10^{-18} \pm 8.92 \times 10^{-19}$	$7.28 \times 10^{-17} \pm 2.06 \times 10^{-17}$	$3.34 \times 10^{-19} \pm 1.34 \times 10^{-19}$
F_{11}	20	$2.03 \times 10^{-2} \pm 9.95 \times 10^{-3}$	$1.85 \times 10^{-5} \pm 1.30 \times 10^{-5}$	$3.29 \times 10^{-8} \pm 2.52 \times 10^{-8}$	$2.98 \times 10^{-8} \pm 1.44 \times 10^{-8}$	$4.76 \times 10^{-9} \pm 3.51 \times 10^{-9}$
	30	$1.04 \times 10^{-1} \pm 3.33 \times 10^{-2}$	$6.60 \times 10^{-4} \pm 3.86 \times 10^{-4}$	$1.17 \times 10^{-4} \pm 1.14 \times 10^{-4}$	$1.92 \times 10^{-7} \pm 8.18 \times 10^{-8}$	$1.41 \times 10^{-6} \pm 1.04 \times 10^{-6}$
	50	$3.54 \times 10^{-1} \pm 7.46 \times 10^{-2}$	$1.60 \times 10^{-1} \pm 1.32 \times 10^{-1}$	$2.81 \times 10^{-2} \pm 1.48 \times 10^{-2}$	$2.13 \times 10^{-3} \pm 2.60 \times 10^{-3}$	$2.43 \times 10^{-3} \pm 1.41 \times 10^{-3}$
	100	$1.63 \pm 2.12 \times 10^{-1}$	$1.90 \times 10^1 \pm 3.61$	$2.74 \pm 6.02 \times 10^{-1}$	$2.85 \times 10^{-1} \pm 4.42 \times 10^{-2}$	$7.30 \times 10^{-1} \pm 3.31 \times 10^{-1}$
F_{12}	20	$8.66 \times 10^{-3} \pm 5.26 \times 10^{-3}$	$1.53 \times 10^{-8} \pm 4.57 \times 10^{-9}$	$1.72 \times 10^{-9} \pm 6.92 \times 10^{-10}$	$9.91 \times 10^{-8} \pm 7.70 \times 10^{-8}$	$2.19 \times 10^{-10} \pm 1.29 \times 10^{-10}$
	30	$3.15 \times 10^{-2} \pm 1.41 \times 10^{-2}$	$3.42 \times 10^{-8} \pm 9.83 \times 10^{-9}$	$3.83 \times 10^{-9} \pm 1.82 \times 10^{-9}$	$2.51 \times 10^{-6} \pm 2.88 \times 10^{-6}$	$4.04 \times 10^{-10} \pm 1.71 \times 10^{-10}$
	50	$6.11 \times 10^{-2} \pm 2.19 \times 10^{-2}$	$8.01 \times 10^{-8} \pm 2.73 \times 10^{-8}$	$9.29 \times 10^{-9} \pm 2.72 \times 10^{-9}$	$3.49 \times 10^{-5} \pm 4.01 \times 10^{-5}$	$7.00 \times 10^{-10} \pm 1.88 \times 10^{-10}$
	100	$2.13 \times 10^{-1} \pm 4.75 \times 10^{-2}$	$2.28 \times 10^{-7} \pm 4.18 \times 10^{-8}$	$4.18 \times 10^{-8} \pm 1.21 \times 10^{-8}$	$3.19 \times 10^{-4} \pm 3.87 \times 10^{-4}$	$1.36 \times 10^{-9} \pm 1.79 \times 10^{-10}$
F_{13}	20	$1.13 \times 10^{-11} \pm 5.95 \times 10^{-12}$	$2.41 \times 10^3 \pm 6.47 \times 10^2$	$6.64 \times 10^2 \pm 4.27 \times 10^2$	$1.82 \times 10^{-12} \pm 0$	$2.97 \times 10^{-12} \pm 1.12 \times 10^{-12}$
	30	$3.79 \times 10^{-11} \pm 6.89 \times 10^{-11}$	$3.14 \times 10^3 \pm 7.71 \times 10^2$	$8.64 \times 10^2 \pm 6.67 \times 10^2$	$1.82 \times 10^{-12} \pm 0$	$3.64 \times 10^{-12} \pm 1.51 \times 10^{-12}$
	50	$4.16 \times 10^2 \pm 3.62 \times 10^2$	$5.56 \times 10^3 \pm 1.31 \times 10^3$	$2.00 \times 10^3 \pm 4.94 \times 10^2$	$2.55 \times 10^{-11} \pm 0$	$3.61 \times 10^{-11} \pm 7.79 \times 10^{-12}$
	100	$6.39 \times 10^2 \pm 5.13 \times 10^2$	$1.07 \times 10^4 \pm 1.92 \times 10^3$	$4.65 \times 10^3 \pm 1.91 \times 10^3$	$1.51 \times 10^{-10} \pm 1.54 \times 10^{-11}$	$1.72 \times 10^{-10} \pm 1.69 \times 10^{-11}$
F_{14}	20	$2.18 \times 10^{-13} \pm 8.21 \times 10^{-14}$	$2.33 \times 10^{-18} \pm 4.63 \times 10^{-18}$	$8.32 \times 10^{-22} \pm 3.84 \times 10^{-22}$	$4.10 \times 10^{-20} \pm 2.42 \times 10^{-20}$	$5.12 \times 10^{-24} \pm 4.43 \times 10^{-24}$
	30	$1.75 \times 10^{-13} \pm 5.71 \times 10^{-14}$	$2.96 \times 10^{-19} \pm 2.96 \times 10^{-19}$	$7.76 \times 10^{-22} \pm 2.30 \times 10^{-22}$	$5.03 \times 10^{-20} \pm 2.42 \times 10^{-20}$	$2.58 \times 10^{-23} \pm 1.31 \times 10^{-23}$
	50	$1.09 \times 10^{-13} \pm 2.63 \times 10^{-14}$	$2.06 \times 10^{-19} \pm 7.89 \times 10^{-20}$	$5.98 \times 10^{-22} \pm 1.54 \times 10^{-22}$	$4.48 \times 10^{-20} \pm 1.43 \times 10^{-20}$	$9.38 \times 10^{-23} \pm 3.19 \times 10^{-23}$
	100	$7.41 \times 10^{-14} \pm 1.27 \times 10^{-14}$	$1.55 \times 10^{-19} \pm 1.19 \times 10^{-19}$	$3.75 \times 10^{-22} \pm 7.30 \times 10^{-23}$	$2.96 \times 10^{-20} \pm 7.23 \times 10^{-21}$	$1.01 \times 10^{-22} \pm 2.71 \times 10^{-23}$
F_{15}	20	$1.17 \times 10^{-11} \pm 3.31 \times 10^{-12}$	$8.51 \times 10^{-18} \pm 3.20 \times 10^{-18}$	$3.74 \times 10^{-20} \pm 1.64 \times 10^{-20}$	$1.46 \times 10^{-18} \pm 8.73 \times 10^{-19}$	$5.55 \times 10^{-23} \pm 4.91 \times 10^{-23}$
	30	$8.80 \times 10^{-12} \pm 2.71 \times 10^{-12}$	$7.08 \times 10^{-18} \pm 2.59 \times 10^{-18}$	$3.28 \times 10^{-20} \pm 1.09 \times 10^{-20}$	$1.78 \times 10^{-18} \pm 8.42 \times 10^{-19}$	$1.81 \times 10^{-22} \pm 1.30 \times 10^{-22}$
	50	$7.55 \times 10^{-12} \pm 1.77 \times 10^{-12}$	$8.89 \times 10^{-18} \pm 2.34 \times 10^{-18}$	$3.14 \times 10^{-20} \pm 8.39 \times 10^{-21}$	$2.39 \times 10^{-18} \pm 7.91 \times 10^{-19}$	$2.29 \times 10^{-21} \pm 7.71 \times 10^{-22}$
	100	$6.85 \times 10^{-12} \pm 1.18 \times 10^{-12}$	$8.41 \times 10^{-18} \pm 1.48 \times 10^{-18}$	$3.19 \times 10^{-20} \pm 6.77 \times 10^{-21}$	$2.47 \times 10^{-18} \pm 5.65 \times 10^{-19}$	$6.92 \times 10^{-21} \pm 1.84 \times 10^{-21}$

TABLE V
WILCOXON RANK-SUM TEST RESULTS OF STA VARI

TABLE VI
COMPARISON OF AVERAGE GRADIENT NORM (GRADNORM) UNDER THE MAXSTALLS TERMINATION CRITERION

Fcn	Dim	GL25	CLPSO	CMA-ES	LSHADE	GWO	WOA	PGA	KLA	ESTA	EXSTA
F_1	20	8.74×10^{-1}	6.49×10^{-1}	2.04×10^{-8}	9.36×10^{-9}	3.12×10^{-1}	2.46	2.91×10^{-8}	2.37×10^{-8}	8.73×10^{-8}	4.38×10^{-9}
	30	1.87×10^{-2}	1.54×10^{-2}	9.06×10^{-1}	1.05×10^{-8}	6.84×10^{-1}	5.14	4.47×10^{-8}	3.62×10^{-8}	1.14×10^{-7}	7.49×10^{-9}
	50	4.04×10^{-2}	4.50×10^{-2}	1.93×10^{-2}	1.15×10^{-8}	1.90×10^{-2}	1.48×10^{-1}	7.23×10^{-8}	6.28×10^{-8}	1.68×10^{-7}	1.47×10^{-8}
	100	1.32×10^{-3}	1.45×10^{-3}	5.48×10^{-2}	1.76×10^{-8}	6.80×10^{-2}	6.17×10^{-1}	7.61	7.61×10^{-7}	2.83×10^{-7}	2.85×10^{-8}
F_2	20	6.44×10^{-6}	3.15×10^{-6}	4.37×10^{-7}	3.40×10^{-7}	1.65×10^{-5}	1.66×10^{-2}	2.83	6.31×10^{-1}	4.10×10^{-6}	7.71×10^{-7}
	30	2.65×10^{-7}	2.65×10^{-7}	1.18×10^{-2}	3.84×10^{-7}	1.69×10^{-6}	1.02×10^{-3}	4.47	1.64×10^{-2}	4.56×10^{-6}	1.00×10^{-6}
	50	2.40×10^{-8}	1.94×10^{-8}	1.18×10^{-6}	4.74×10^{-7}	2.50×10^{-7}	9.88×10^{-3}	5.07	1.92×10^{-6}	7.87×10^{-6}	1.45×10^{-6}
	100	3.37×10^{-9}	4.64×10^{-9}	1.44×10^{-8}	9.80×10^{-7}	5.85×10^{-8}	4.39×10^{-5}	6.04×10^{-7}	2.76×10^{-6}	1.70×10^{-5}	3.10×10^{-6}
F_3	20	2.02×10^{-2}	1.88×10^{-2}	1.87×10^{-2}	1.66×10^{-2}	1.99×10^{-2}	1.30×10^{-2}	2.11	1.85×10^{-2}	1.64×10^{-5}	9.74×10^{-6}
	30	3.40×10^{-2}	3.09×10^{-2}	2.37×10^{-2}	2.21×10^{-2}	2.42×10^{-2}	2.11×10^{-2}	2.22×10^{-2}	2.28×10^{-2}	3.18×10^{-5}	1.68×10^{-5}
	50	4.93×10^{-2}	5.59×10^{-2}	3.47×10^{-2}	2.98×10^{-2}	3.59×10^{-2}	3.24×10^{-2}	1.51×10^{-2}	3.05×10^{-2}	5.00×10^{-5}	2.51×10^{-5}
	100	1.32×10^{-3}	1.49×10^{-3}	6.96×10^{-2}	4.30×10^{-2}	8.05×10^{-2}	5.08×10^{-2}	4.16×10^{-2}	4.35×10^{-2}	1.40×10^{-4}	6.11×10^{-5}
F_4	20	1.79×10^{-2}	1.72×10^{-2}	5.24×10^{-3}	1.07×10^{-8}	7.37×10^{-3}	1.03×10^{-1}	1.11×10^{-8}	1.08×10^{-8}	2.88×10^{-8}	1.19×10^{-8}
	30	4.34×10^{-2}	4.22×10^{-2}	1.17×10^{-2}	1.02×10^{-8}	1.70×10^{-2}	1.59×10^{-1}	1.49×10^{-8}	1.28×10^{-8}	3.80×10^{-8}	1.37×10^{-8}
	50	1.04×10^{-1}	1.06×10^{-1}	4.37×10^{-2}	9.90×10^{-9}	4.78×10^{-2}	1.56×10^{-1}	1.75×10^{-1}	1.57×10^{-8}	4.35×10^{-8}	1.41×10^{-8}
	100	3.39×10^{-1}	3.56×10^{-1}	1.44×10^{-1}	1.02×10^{-8}	1.74×10^{-1}	1.41×10^{-2}	1.42×10^{-1}	2.48×10^{-8}	7.06×10^{-8}	1.61×10^{-8}
F_5	20	1.01	1.12	1.37	6.45×10^{-10}	1.17	1.45	1.05×10^{-9}	9.90×10^{-10}	2.84×10^{-4}	1.25×10^{-7}
	30	8.89×10^{-1}	1.17	1.15	1.25×10^{-9}	9.33×10^{-1}	1.27	1.69×10^{-9}	2.86×10^{-3}	2.07×10^{-4}	1.35×10^{-6}
	50	8.88×10^{-1}	8.30×10^{-1}	8.36×10^{-1}	9.90×10^{-1}	8.84×10^{-1}	2.58×10^{-3}	1.32×10^{-1}	2.88×10^{-5}	4.50×10^{-4}	5.50×10^{-6}
	100	5.47×10^{-1}	5.20×10^{-1}	5.48×10^{-1}	6.42×10^{-1}	5.85×10^{-1}	3.55×10^{-6}	3.40×10^{-6}	0.00	2.80×10^{-4}	3.96×10^{-7}
F_6	20	3.28×10^{-2}	5.32×10^{-3}	1.32×10^{-3}	7.96×10^{-8}	7.03×10^{-2}	1.96×10^{-2}	5.76×10^{-2}	8.15	2.98×10^{-7}	3.62×10^{-8}
	30	4.16×10^{-7}	1.40×10^{-10}	8.30×10^{-3}	1.67×10^{-7}	3.45×10^{-3}	3.34×10^{-2}	6.24×10^{-3}	5.40	6.72×10^{-7}	1.09×10^{-7}
	50	1.86×10^{-12}	2.22×10^{-12}	3.69×10^{-4}	3.68×10^{-3}	1.25×10^{-5}	6.39×10^{-2}	1.35×10^{-4}	3.12	8.01×10^{-7}	2.28×10^{-7}
	100	6.78×10^{-16}	1.11×10^{-17}	1.88×10^{-6}	1.22×10^{-5}	3.30×10^{-5}	1.11×10^{-4}	8.41×10^{-5}	3.45	9.37×10^{-6}	1.25×10^{-6}
F_7	20	5.00×10^{-1}	7.00×10^{-1}	6.73×10^{-1}	7.52×10^{-1}	6.19×10^{-1}	4.15×10^{-1}	4.48×10^{-1}	6.27	2.15×10^{-5}	9.92×10^{-6}
	30	1.27×10^{-2}	9.66×10^{-1}	1.15×10^{-2}	1.27×10^{-2}	1.10×10^{-2}	9.18×10^{-1}	1.25×10^{-2}	1.04	5.70×10^{-5}	2.45×10^{-5}
	50	1.80×10^{-2}	2.50×10^{-2}	2.39×10^{-2}	2.73×10^{-2}	2.19×10^{-2}	2.03×10^{-2}	2.40×10^{-2}	2.39	1.99×10^{-4}	7.77×10^{-5}
	100	5.67×10^{-2}	5.98×10^{-2}	6.38×10^{-2}	7.21×10^{-2}	5.42×10^{-2}	5.74×10^{-2}	5.97×10^{-2}	6.48	1.30×10^{-3}	4.28×10^{-4}
F_8	20	6.65×10^{-2}	4.65×10^{-2}	1.41×10^{-5}	1.23×10^{-5}	1.68×10^{-2}	8.81	9.50×10^{-3}	1.82×10^{-5}	3.94×10^{-5}	2.16×10^{-5}
	30	2.16×10^{-3}	1.97×10^{-3}	4.43×10^{-5}	3.90×10^{-5}	2.97×10^{-2}	1.22×10^{-1}	1.48×10^{-1}	6.63×10^{-5}	1.56×10^{-4}	7.17×10^{-5}
	50	8.51×10^{-3}	8.00×10^{-3}	5.76×10^{-4}	5.44×10^{-4}	3.51×10^{-1}	1.47×10^{-1}	1.67	2.55×10^{-2}	1.27×10^{-3}	6.75×10^{-4}
	100	5.03×10^{-4}	5.72×10^{-4}	2.46×10^{-2}	2.28×10^{-2}	3.43×10^{-1}	2.01×10^{-1}	1.29×10^{-2}	9.64	2.52×10^{-2}	2.51×10^{-2}
F_9	20	5.36×10^{-3}	3.88×10^{-3}	2.49×10^{-2}	4.01×10^{-8}	1.32×10^{-2}	9.06×10^{-1}	2.27×10^{-1}	1.70×10^{-7}	1.71×10^{-7}	3.35×10^{-8}
	30	3.46×10^{-4}	4.24×10^{-4}	1.10×10^{-3}	5.53×10^{-8}	3.22×10^{-2}	3.29×10^{-1}	3.13×10^{-1}	4.95×10^{-1}	3.01×10^{-7}	5.10×10^{-8}
	50	2.46×10^{-5}	2.74×10^{-5}	8.42×10^{-3}	1.19×10^{-7}	1.94×10^{-3}	1.43×10^{-3}	5.30×10^{-2}	9.73	5.80×10^{-7}	1.10×10^{-7}
	100	3.25×10^{-6}	3.81×10^{-6}	1.42×10^{-5}	3.26×10^{-7}	1.68×10^{-4}	1.29×10^{-4}	4.97×10^{-3}	1.41	1.25×10^{-6}	3.96×10^{-7}
F_{10}	20	3.76×10^{-4}	3.59×10^{-4}	5.06×10^{-8}	2.42×10^{-8}	1.90×10^{-3}	9.17	1.04×10^{-7}	1.03×10^{-7}	2.02×10^{-7}	1.77×10^{-8}
	30	2.59×10^{-5}	2.64×10^{-5}	6.43×10^{-8}	2.70×10^{-8}	1.90×10^{-4}	6.07×10^{-1}	1.41×10^{-7}	1.61×10^{-7}	2.75×10^{-7}	2.96×10^{-8}
	50	2.22×10^{-6}	2.73×10^{-6}	1.10×10^{-5}	2.94×10^{-8}	2.56×10^{-5}	3.26×10^{-2}	2.27×10^{-7}	2.23×10^{-7}	4.11×10^{-7}	4.94×10^{-8}
	100	3.33×10^{-7}	4.35×10^{-7}	1.72×10^{-6}	4.71×10^{-8}	6.27×10^{-6}	5.01×10^{-3}	3.22×10^{-5}	4.40×10^{-7}	5.47×10^{-7}	9.44×10^{-8}

TABLE VII
STATISTICAL RESULTS OF DIFFERENT ALGORITHMS ON THE SNL PROBLEM

Index	GL25	CLPSO	CMA-ES	LSHADE	GWO	WOA	PGA	KLA	ESTA	EXSTA
Best	2.26×10^{-1}	5.09×10^{-4}	1.48×10^{-28}	1.92×10^{-8}	2.25×10^{-1}	4.30×10^{-2}	1.15×10^{-2}	6.41×10^{-6}	6.41×10^{-15}	1.87×10^{-18}
Mean	6.07×10^{-1}	7.30×10^{-4}	1.62×10^{-28}	1.28×10^{-6}	2.72×10^{-1}	4.16×10^{-1}	1.68	4.72×10^{-5}	4.58×10^{-14}	1.51×10^{-17}
Worst	1.07	1.90×10^{-3}	1.79×10^{-28}	4.44×10^{-6}	3.36×10^{-1}	9.39×10^{-1}	5.27	1.60×10^{-4}	1.10×10^{-13}	4.11×10^{-17}
Time(s)	82.6	75.1	440.3	86.2	78.4	65.4	68.6	87.9	84.5	68.3

on imitation learning or behavioral cloning, which lack the necessary mechanisms to ensure optimality, and they tend to stagnate at stable points [17].

2) *Terminate at maxFEs*: This section evaluates the proposed effective STAs against other algorithms for the sensor network localization (SNL) problem, using a fixed budget of maxFEs. Following the formulation in [11], the SNL problem is described as follows. Given m anchor points $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^d$ ($d = 2$ in this study) and a radio range r_d , let $N_x = \{(i, j) : \|\mathbf{x}_i - \mathbf{x}_j\| = d_{ij} \leq r_d\}$ denote the set of measurable sensor-to-sensor distances and $N_a = \{(i, k) : \|\mathbf{x}_i - \mathbf{a}_k\| = \bar{d}_{ik} \leq r_d\}$ denote the set of measurable sensor-to-anchor distances. The goal is to determine the positions of n distinct sensor points $\mathbf{x}_i \in \mathbb{R}^2$, for $i = 1, \dots, n$, that satisfy the distance constraints for all pairs in N_x and N_a as follows:

$$\|\mathbf{x}_i - \mathbf{x}_j\|^2 = d_{ij}^2 \quad \forall (i, j) \in N_x \quad (13a)$$

$$\|\mathbf{x}_i - \mathbf{a}_k\|^2 = \bar{d}_{ik}^2 \quad \forall (i, k) \in N_a. \quad (13b)$$

Employing the nonlinear least-squares method, the SNL problem can be reformulated as the following nonconvex optimization problem:

$$\min \sum_{(i,j) \in N_x} (\|\mathbf{x}_i - \mathbf{x}_j\|^2 - d_{ij}^2)^2 + \sum_{(i,k) \in N_a} (\|\mathbf{x}_i - \mathbf{a}_k\|^2 - \bar{d}_{ik}^2)^2. \quad (14)$$

In the experiments, the SNL problem with 100 sensors is considered. The maxFE is set to $10^4 \times D$, where $D = 2n = 200$. The search domain is restricted to the interval $[0, 1]$. Each algorithm is independently executed 30 times. The statistical results are summarized in Table VII, while the best solution obtained by each algorithm is illustrated in Fig. 6, where green circles denote the true sensor positions and red asterisks represent the estimated positions.

It is observed that CMA-ES, ESTA, and EXSTA consistently achieve the global best solution with high precision, but CMA-ES requires more computational time. That is to say, the tradeoff between computational cost and efficiency still exists. It is observed that ESTA occasionally incurs high computational time due to its parameter settings, while EXSTA's computation time is competitive with that of its competitors. Notably, EXSTA achieves a very high accuracy while requiring significantly less computational time. In contrast, GWO performs poorly on this problem.

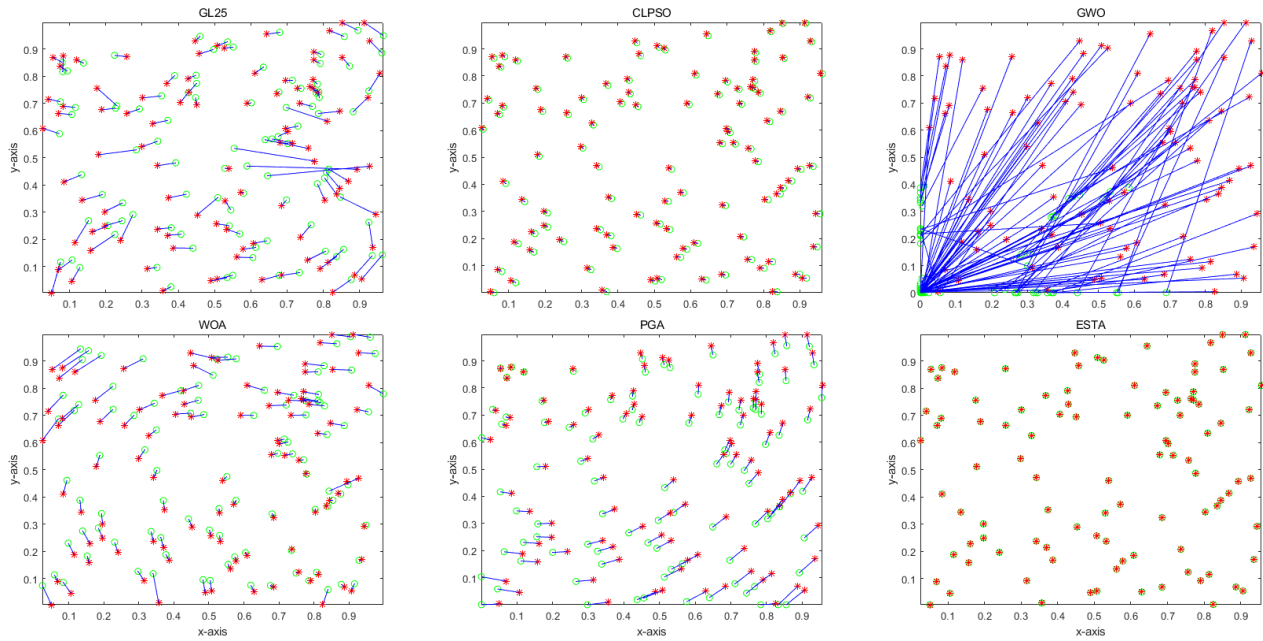


Fig. 6. Best results obtained by different algorithms for the SNL problem.

V. CONCLUSION

This study proposes efficient STAs with guaranteed optimality for system engineering optimization. First, novel TTs based on predictive models are designed to generate more promising candidate solutions, thereby significantly accelerating convergence. Second, parameter control strategies are developed to enhance search efficiency. Finally, a dedicated termination criterion is designed to enable convergence to optimality. Comparative studies with both well-established and recently developed metaheuristics demonstrate the effectiveness and superiority of the proposed approach. Notably, even with a relatively small sample size ($SE = 30$), the proposed ESTA and EXSTA are capable of achieving high-precision solutions close to the global optimum. It is also observed that ESTA may occasionally converge to local optima on low-dimensional Griewank functions, where near-global solutions are located far from the true global optimum, making it difficult to escape local minima.

It should be noted that the proposed ESTA occasionally yields locally optimal results on the low-dimensional Griewank function, as the near-global solution in the Griewank function is located far from the true global solution. In such cases, the proposed ESTA faces difficulty in escaping local minima. Nevertheless, the proposed approach, along with the majority of existing studies on STA, primarily focuses on individual- and sampling-based methods. Considering that the population-based algorithms can explore different regions of the search space simultaneously, massively reducing the chance of getting trapped in a single local optimum, in the future, population-based STA will be further explored to enhance its global search capabilities. In the meantime, more real-world applications will be investigated to further validate the effectiveness and applicability of the proposed method.

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